

README

Project Title

Arduino Temperature Monitoring System

Project Overview

This project demonstrates real-time temperature monitoring using an Arduino simulation environment. The program reads temperature values, displays them on a 16x2 I2C LCD, and logs the readings to the Serial Monitor.

Objective

Read temperature values, display them on the LCD, and continuously log them to the Serial Monitor.

Software & Libraries

- Arduino IDE
- LiquidCrystal_I2C
- OneWire
- DallasTemperature

Arduino Sketch

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h> #include
<OneWire.h>
#include <DallasTemperature.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
#define ONE_WIRE_BUS 2

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

void setup()
{ Serial.begin(9600);
  sensors.begin(); lcd.init();
  lcd.backlight();
  lcd.setCursor(0,0);
  lcd.print("Temperature");
}

void loop()
{ sensors.requestTemperatures();
  float tempC = sensors.getTempCByIndex(0);

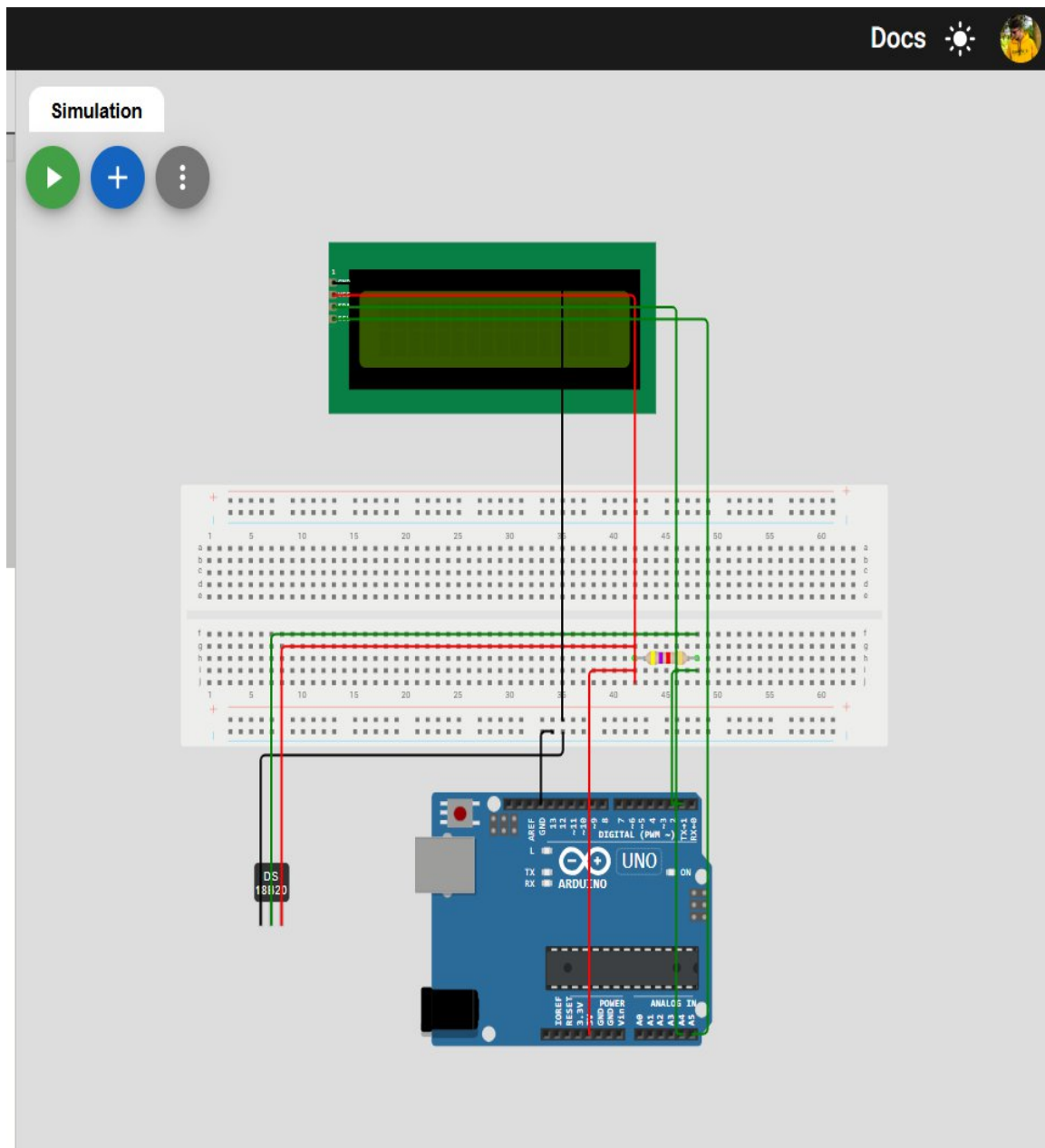
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Temperature");

  lcd.setCursor(0,1); lcd.print(tempC);
  lcd.print((char)223); lcd.print("C");

  Serial.print("Temperature: ");
  Serial.print(tempC); Serial.println("
C");

  delay(1000);
}
```

Simulation Circuit



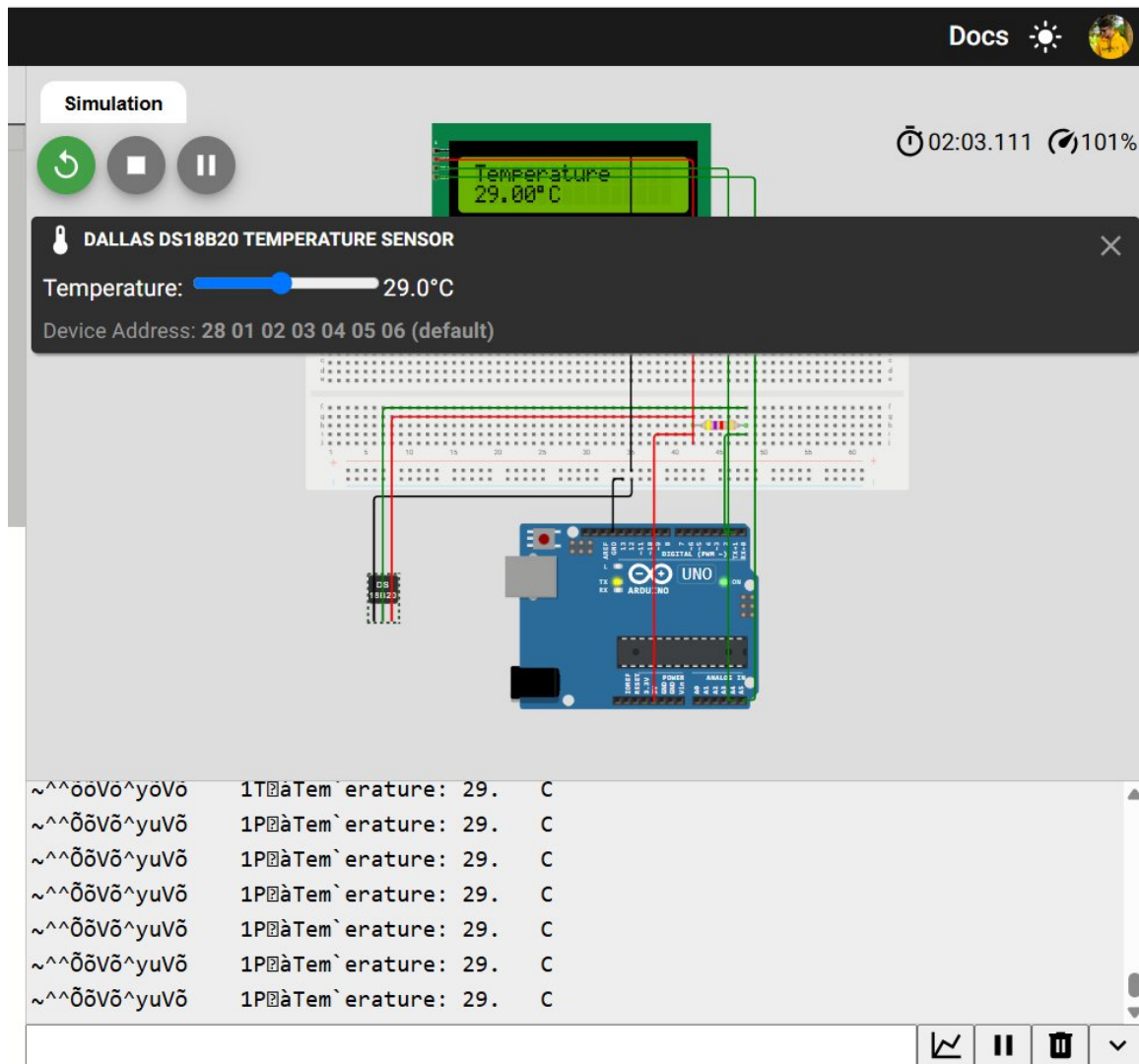
Working Principle

The program initializes the LCD and temperature sensor, reads the current temperature every second, displays it on the LCD, and prints it to the Serial Monitor.

Output

LCD: Temperature / 29.00°C

Serial Monitor: Temperature: 29.00C



Demo Video

https://www.linkedin.com/posts/pavan-k-382a33372_alfidotech-embeddedsystems-arduino-ugcPost-7486653129724317696-_7Kx/?utm_source=share&utm_medium=member_android&rcm=ACoAAFxYYVIBecldMJrji7ICiaCcWsh52kzoq08

Deliverables

- Arduino Sketch (.ino)
- Simulation Screenshots
- Demo Video (30–60 seconds)
- README PDF

Conclusion

The project successfully demonstrates temperature monitoring and serial logging in a simulation environment.